

Transient response performances and control strategy of porous volumetric solar receivers with different heat transfer fluids

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ABSTRACT

The transient behavior of solar receivers significantly impacts start-up, shutdown, and fluid outlet temperature stability. This study investigates the transient behaviors of porous volumetric solar receivers using various heat transfer fluids under step increases and actual solar radiation. The porous domain is reconstructed using the unit cell structure of Weaire-Phelan. The implementation of the solar radiation process, coupled flow and heat transfer, as well as control algorithm leverages Monte Carlo ray tracing, pore-scale simulation, and PID control, respectively. Results demonstrate that air and molten salt porous receivers display concave and s-shaped response curves when exposed to a step increase in solar radiation. Molten salt's higher heat capacity compared to that of air causes a delayed response to solar radiation variations with an average time constant of 45.9 s, whereas the air receiver's time constant is only 7.0 s. Furthermore, the time constant increases by 129 % as the mass flow rate decreases from 9×10^{-5} to 3×10^{-5} kg/s. The PID algorithm effectively reduces temperature fluctuation amplitudes caused by solar radiation variation. The maximum temperature offsets for air and molten salt porous receivers are 5.9 K and 31.4 K, respectively.

1. Introduction

Solar energy is widely regarded as the paramount source for future human societies. Concentrating Solar Power (CSP) technology offers an efficient mechanism for the high-temperature and large-scale utilization of solar energy [1,2]. However, the diurnal and intermittent nature of solar energy poses challenges to the stable operation of CSP systems [3]. The transient performance of the solar receiver, a core component responsible for solar-to-thermal energy conversion, plays a crucial role in determining the temperature of the heat transfer fluid and the overall thermal efficiency of the system. Furthermore, the high temperature and steep temperature gradient significantly affects the long-term safe operation of the receivers [4].

Volumetric absorption stands out as a promising approach for efficiently absorbing and converting solar radiation by mitigating thermal radiation loss [5]. Nevertheless, the complex flow and coupled heat transfer processes within porous media present challenges in precisely characterizing thermal performances, especially concerning the transient response performance of porous volumetric solar receivers.

In comparison to the volume-averaging simulation method, pore-scale simulation proves to be an effective tool for analyzing the

intricate and detailed energy transfer processes within porous structures. Zhu and Xuan [6] conducted simulations on packed bed porous media, investigating the influence of geometrical parameters on radiation propagation, pressure drop, and thermal efficiency. A hybrid geometry with reduced front surface temperature was proposed. Du et al. [7,8] employed X-ray computed tomography to reconstruct porous geometry and developed image processing procedures to adjust geometrical parameters such as porosity and pore size. Monte Carlo ray tracing and pore-scale simulation methods were applied to examine the steady-state flow and thermal performance of porous volumetric solar receivers under intensified solar irradiance. According to the spectrally selective absorption concept, Du et al. [9] further designed a porous receiver with molten salt as heat transfer fluid. The flow and thermal performances were compared to the air porous receivers with pore-scale simulation method, and the low flow resistance and high thermal efficiency were characterized. Navalho and Pereira [10] simulated the entire solar radiation transfer process, encompassing solar energy concentration in a parabolic dish and radiation transfer within the porous foam. Chen et al. [11] investigated influences of the optical configurations on the thermal performance of the receiver. The decreasing absorptivity and spectrally selective absorption designs were beneficial for improving the energy conversion efficiency. Additionally, based on the adjoint algorithm and

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Nomenclature

c_p	specific heat (J/(kg•K))
e	temperature difference (K)
K_p	proportional coefficient in PID controller
$\dot{m}$	mass flow rate (kg/s)
p	pressure (Pa)
S_r	energy source term (W)
T	temperature (K)
T_{sampling}	derivative coefficient in PID controller
T_d	derivative coefficient in PID controller
T_i	integral coefficient in PID controller
u	velocity (m/s)
x	coordinate (m)

Greek symbols

λ	thermal conductivity (W/(m•K))
μ	dynamic viscosity (Pa•s)
ρ	density (kg/(m ³))
τ_t	thermal tortuosity

Subscripts

f	fluid
s	solid

pore-scale reconstruction, Liu et al. [12] and Fan et al. [13] optimized the radiative heat transfer process and flow resistance characteristics of the porous receivers, respectively. An improvement in absorption efficiency of about 20 % and a maximum pressure drop reduction of 43 % were achieved, respectively.

Several transient experimental tests on porous volumetric solar receivers were conducted in the early studies [14–16]. However, numerical simulations provide effective tools for a detailed examination of transient thermal performance. Wu and Wang [17] developed a transient model for porous volumetric solar receivers, considering sudden heat flux, sudden loss of heat flux, and heat flux step changes. The results demonstrated that the reaction time of the receiver's outlet temperature fell within the range of tens of seconds, and a smooth change in temperature and pressure drop occurred even during sudden variations in solar radiation. In a study by Wang et al. [18], the influences of fluid type and thermal conductivity of porous material on the transient performance of porous receivers were investigated. It was observed that fluid type had a slight impact on the inlet temperature of the porous skeleton but significantly affected the heat transfer process of the outlet porous temperature. Ribeiro and Lemos [19] utilized the volume-averaging method with a local thermal non-equilibrium model to explore the transient temperature response of porous receivers. The findings indicated that low porosity and decreasing Darcy number

facilitated effective heat exchange between phases, leading to a quicker stabilization of outlet temperature. Sharma and Talukdar [20] approximated various operational scenarios involving linearly changing solar flux, including start-up, shutdown, and cloud passage for porous receivers. The temperature variations over time were recorded, revealing a delayed response at the rear part of the porous receiver. Additionally, Cagnoli et al. [21] proposed a multi-scale analysis method, spanning from micro-scale to macro-scale, for porous volumetric solar receivers. The simulation of the cloud passage process involved imposing zero solar heat flux for 20 s. Macro-scale numerical results indicated an immediate response of the front surface of the porous receiver to variations in solar heat flux, with a rapid temperature change rate of approximately 15 K/s. Similar quick temperature variation rates exceeding 20 K/s were also observed by Reddy and Nataraj [22].

On the other hand, diverse control strategies have been employed to regulate the temperature of solar receivers and stabilize transient performance in various solar thermal applications. This involves manipulating the heliostat field, receiver aperture size, and mass flow rate of the heat transfer fluid. In a study by Popp et al. [23], a comparison was made between model predictive control (MPC) and a PID-based control method for regulating the temperature of a molten salt solar receiver by adjusting fluid mass flow. Results across various cases indicated that MPC exhibited superior performance in terms of temperature deviation and compliance with safety standards. However, thermal efficiencies of the receiver were nearly identical when using both MPC and PID control methods. Abuseada and Ozalp [24] employed both PID and MPC methods to regulate the apertures of a cavity solar receiver, aiming to stabilize its temperature under fluctuating solar radiation. Experimental findings demonstrated that both control methods effectively maintained the receiver's temperature within 5°C of the set point under actual solar irradiation. To address the inherent delay characteristic of PID control, Lu et al. [25] and Wang et al. [26] applied feedforward-feedback control strategies in parabolic trough and solar tower receivers, respectively. Lu et al. [25] introduced a hybrid control strategy incorporating model-based feedforward and PI feedback controllers. This strategy compensated for various disturbances such as changes in solar position, ambient temperature, and wind speed, resulting in an approximate 0.5–1 % increase in thermal efficiency. Wang et al. [26] integrated an artificial neural network (ANN)-based feedback controller with the PID feedback controller to enhance the response speed of the solar tower receiver to variations in solar irradiation. The hybrid control strategy effectively limited outlet temperature fluctuations to 5°C under actual Direct Normal Irradiance (DNI) data, whereas this value exceeded 30°C in the PID-controlled system.

Through the preceding literature reviews, it becomes evident that the transient behaviors of porous receivers with various heat transfer fluids remain unexplored, and a comprehensive understanding of the transient performance of porous volumetric solar receivers under actual solar irradiance is yet to be attained. Additionally, the response characteristics of porous receivers to controlling algorithms require further

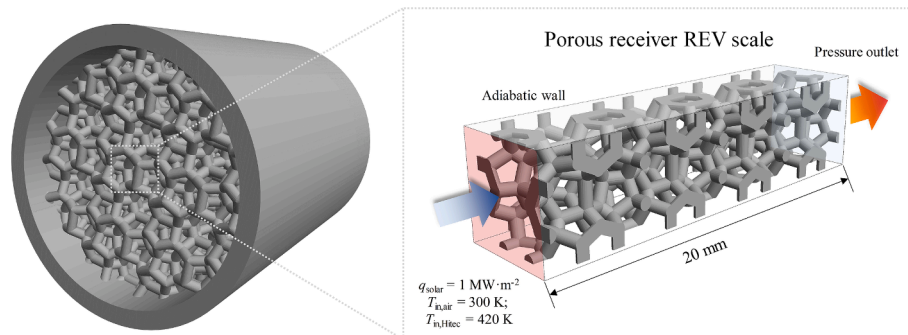


Fig. 1. Illustration of porous receiver with Weaire-Phelan structure as unit cell.

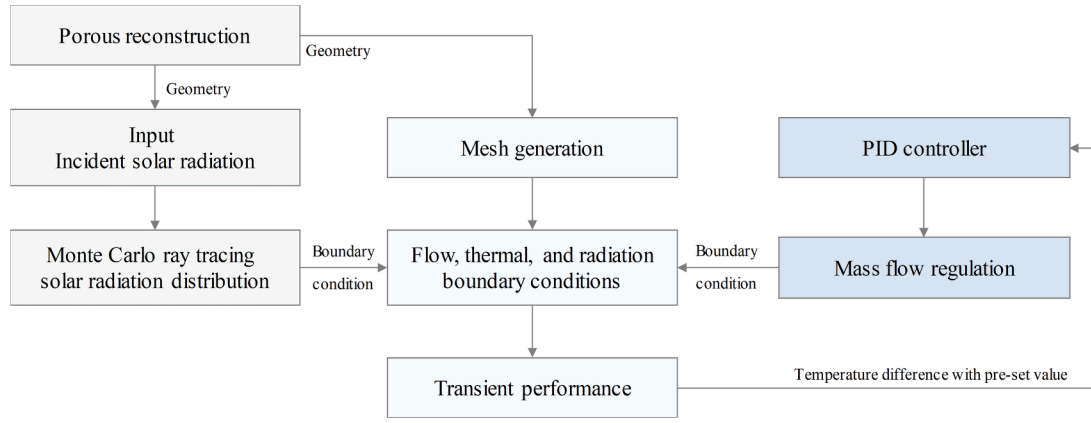


Fig. 2. Transient performance evaluation and PID regulation procedures.

investigation. Notably, prior numerical tools predominantly utilized the volume-averaging simulation method, while a demand for transient studies relying on pore-scale structures, capable of capturing finer details, persists. Consequently, this study aims to comprehensively explore the transient response performances of porous volumetric solar receivers utilizing different heat transfer fluids, both with and without regulation. The porous structure reconstruction, solar radiation calculation, and the PID control methods are illustrated in the Section 2. The pore-scale simulation method for the porous receivers with different heat transfer fluids is presented in Section 3. Transient response performances under the step change and actual solar flux as well as the influence of controlling algorithm are analyzed in Section 4.

2. Methodology

2.1. Weaire-Phelan porous model reconstruction

Pore-scale simulation is conducted by directly reconstructing a three-dimensional porous structure, as shown in Fig. 1. The Weaire-Phelan porous model is chosen and reconstructed using the periodic unit cell reconstruction method, mainly to mitigate the substantial computational resource demands during transient simulation.

Firstly, the unit cell of Weaire-Phelan structure including its vertices' positions and connection information is obtained based on the built-in model of the software Surface Evolver. Then, the geometrical data of the Weaire-Phelan structure is imported to the CAD design software such as Inventor to create the three-dimensional entity. The 3D sketch is drawn by the VB macros and the porous skeleton is formed using the Insert Frame function. Finally, the Miter function is applied to mend the gaps between the skeletons and the porosity of the porous structure is controlled by changing the skeleton thickness. The porosity, pore diameter, strut thickness, and specific surface area of the reconstructed Weaire-Phelan porous model used for the following simulation are 0.8, 2.3 mm, 0.55 mm, and 1.4 mm^{-1} , respectively.

2.2. Monte Carlo ray tracing method

The solar radiation penetration and solar flux distribution inside the porous structure significantly influence the flow and heat transfer performance of the receiver. Before launching the Monte Carlo ray tracing method, the initial state of the solar radiation is determined according to the previous study of Qiu et al. [27]. It is shown that the incident angle varies from 0 to about 45° . Therefore, the solar flux distribution inside the receiver is a weighted average of distributions under different incident angles.

When considering the solar radiation penetration inside the receiver, the MCRT method is applied due to its flexibility to the complex three-dimensional structures. The basic concept of MCRT method is to

simulate the radiation transfer process by statically analyze the behaviors of a large number of independent photons. Based on the geometrical optics assumption, the behaviors of the photons such as reflection, absorption, and transmission can be captured. The reflections in the interface of the porous structure and the side wall of the receiver are assured to be specular and the absorption of solar radiation only happens at the porous surface.

2.3. PID control method

PID is a commonly used controller in various fields in industry. In order to stabilize the thermal performance of the solar receiver under the changing weather condition, the PID control method is applied. The basic equation showing the PID algorithm is shown by Equation(1). In the equation, $\dot{m}$ represents the mass flow rate in the inlet of the receiver and it is the key parameter to regulate for adjusting the outlet temperature. e is denoted as the temperature difference between the current outlet temperature and the set temperature. K_p , T_i , and T_d are the coefficients to scale the proportional, integral, and derivative controllers, respectively.

$$\dot{m}(t+1) = \dot{m}(t) - \left[K_p e(t+1) + \frac{K_p T_{\text{sampling}}}{T_i} \sum_{n=0}^K e(n) + \frac{K_p T_d}{T_{\text{sampling}}} (e(t+1) - e(t)) \right] \quad (1)$$

The Zeigler-Nichols closed-loop method is adopted to tune the aforementioned PID parameters. A small disturbance of the outlet temperature is induced by changing the set temperature. Then, the proportional gain is adjusted until the oscillation presents a constant amplitude, which derives the critical gain and period of oscillation. Finally, the tuned PID parameters can be determined according to Ref. [26], and the evaluation process is illustrated in Fig. 2.

3. Numerical modeling

3.1. Governing equations

Inside the porous receiver, the fluid flow, heat conduction, convective heat transfer, and thermal radiation transfer processes should be solved simultaneously. The governing equations with transient terms including continuity, momentum, and energy conservations equations are presented as follows.

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_j} (\rho u_j) = 0 \quad (2)$$

Table 1
Convective heat transfer coefficient correlations of porous volumetric solar receivers.

Materials	Density kg/m ³	Viscosity Pa·s	Specific heat J/(kg·K)	Thermal conductivity W/(m ² ·K)
Silicon carbide [30]	3200	/	750	80
Stainless steel [31]	7090	/	500	21.5
Air [30]	Ideal-gas law	Sutherland law	Equation(6)	Equation(7)
Hitec [32]	−0.7337T+2280.22	Equation(8)	1560	Equation(9)

$$\frac{\partial u_i}{\partial t} + \frac{\partial}{\partial x_j} (\rho u_i u_j) = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left(\mu \frac{\partial u_i}{\partial x_j} \right) (i = 1, 2, 3) \quad (3)$$

$$\frac{\partial (\rho T_f)}{\partial t} + \frac{\partial}{\partial x_j} (\rho u_j T_f) = \frac{\partial}{\partial x_j} \left(\frac{\lambda_f}{c_p} \frac{\partial T_f}{\partial x_j} \right) \quad (4)$$

$$\frac{\partial (\rho_s c_p T_s)}{\partial t} = \frac{\partial}{\partial x_j} \left(\lambda_s \frac{\partial T_s}{\partial x_j} \right) + S_r \quad (5)$$

In the equations, u , p , and T stand for velocity, pressure, and temperature, which are the parameters to be solved. The bulk material properties of the fluid and porous skeleton are applied where ρ , μ , c_p , and λ represent density, viscosity, specific heat, and thermal conductivity, respectively. S_r is the energy source term applied to the porous skeleton due to solar and thermal radiation. Thermal radiation heat transfer occurs on the surface of porous media which is determined by discrete ordinates (DO) model. The Monte Carlo ray tracing method to determine the solar radiative heat flux have been validated in the previous studies

[7,28]. The numerical model to describe the flow and heat transfer within the porous receivers have been previously checked, which can be referenced from Ref. [7,8,29]. Notably, the numerical method presented above can only reflect the transient behaviors of the porous structure without considering the thermal inertia of other structures, such as fluid channels.

3.2. Boundary conditions and thermophysical properties

In the inlet of the receiver, the mass-flow-inlet boundary condition is applied, and the mass flow of the fluid is the essential factor to regulate the outlet temperature of the receivers. The constant inlet temperature of air and Hitec molten salt is set to 300 K and 420 K, respectively. Hitec molten salt, a eutectic mixture of potassium nitrate, sodium nitrite, and sodium nitrate in the proportions of 53 %, 40 %, and 7 % by weight, respectively, is widely used in CSP systems. The incident solar flux distribution is also assumed to constant and the radiant intensity is

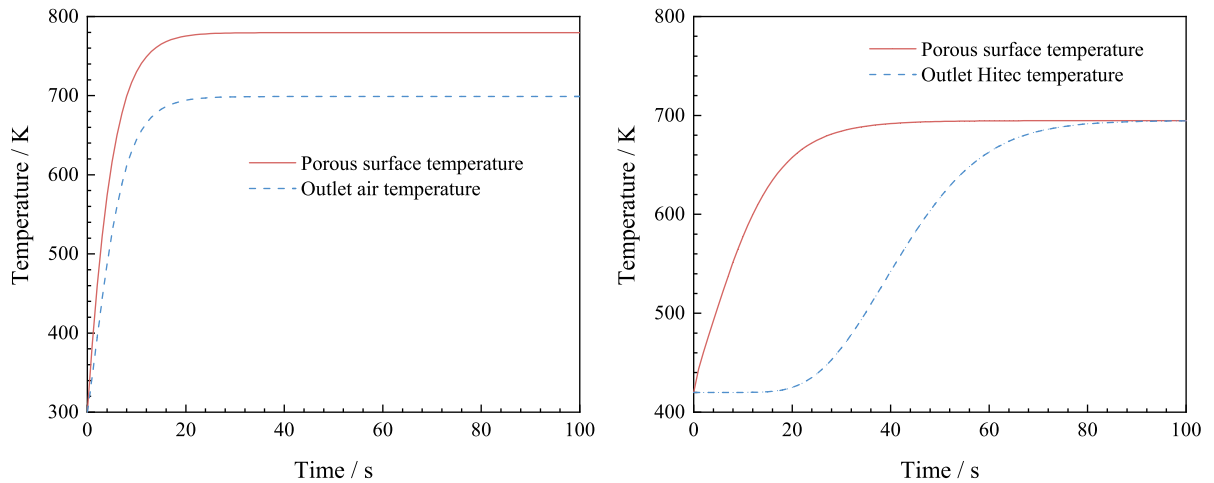


Fig. 3. Temperature variation of porous skeleton and outlet fluid of (a) air and (b) Hitec receivers ($\dot{m}=5 \times 10^{-5}$ kg/s).

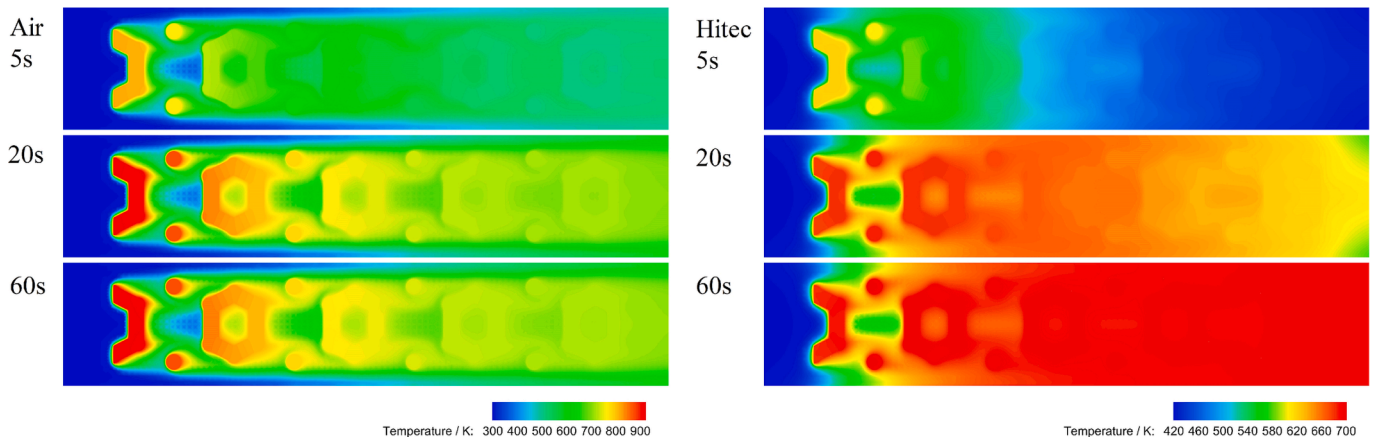


Fig. 4. Temperature fields porous receivers with air and Hitec as heat transfer fluids.

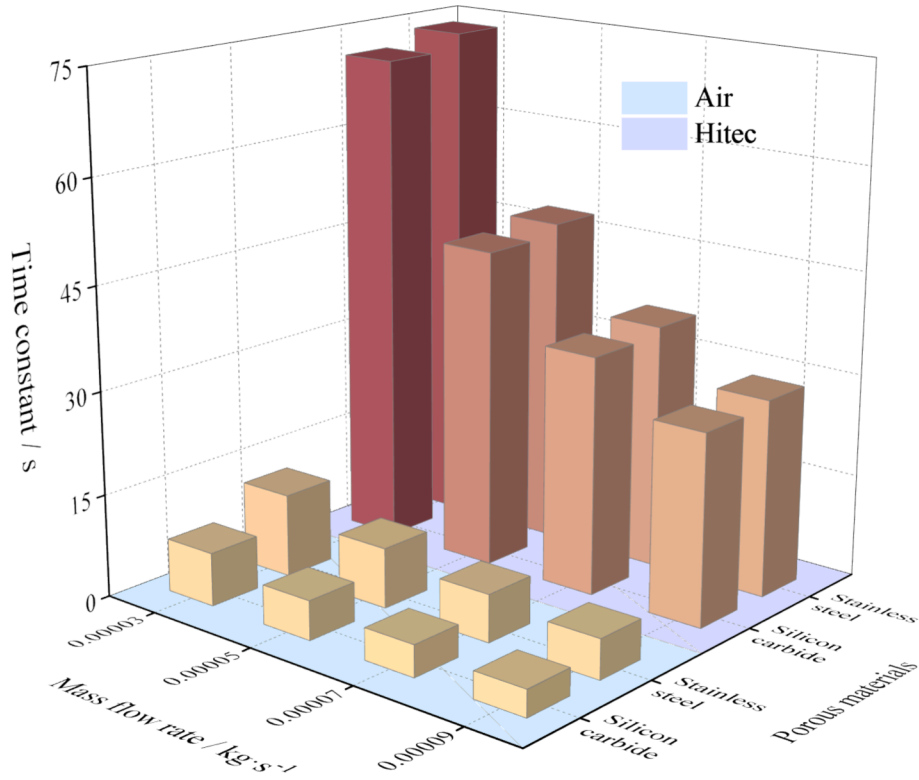


Fig. 5. Time constant of various porous receivers with different fluids, porous materials, and inlet mass flow.

variable. In the outlet of the receiver, the pressure-outlet boundary condition is applied. In the interface of the fluid and porous skeleton, the non-slip and coupled wall boundary condition is used. The emissivity of the porous skeleton is fixed to 0.95. The side walls of the flow channel do not participate in the thermal radiation heat exchange.

$$\lambda_{\text{Air}} = 1.52 \times 10^{-11} T^3 - 4.86 \times 10^{-8} T^2 + 1.02 \times 10^{-4} T - 3.93 \times 10^{-3} \quad (7)$$

$$\mu_{\text{Hitec}} = \begin{cases} -1.742173 \times 10^{-6} T^3 + 2.27615 \times 10^{-3} T^2 \\ -0.991437 T + 143.9826, & 420 \text{ K} \leq T < 440 \text{ K} \\ -7.2058 \times 10^{-9} T^3 + 1.08225 \times 10^{-5} T^2 \\ -5.4754 \times 10^{-3} T + 0.93845, & 440 \text{ K} \leq T < 500 \text{ K} \\ 8.507 \times 10^{-13} T^4 - 2.4331 \times 10^{-9} T^3 + 2.6275 \times 10^{-6} T^2 \\ -1.2768 \times 10^{-3} T + 0.23816, & 500 \text{ K} \leq T \leq 800 \text{ K} \end{cases} \quad (8)$$

$$\lambda_{\text{Hitec}} = \begin{cases} -1.863 \times 10^{-8} T^3 + 2.551 \times 10^{-5} T^2 - 1.176 \times 10^{-2} T + 2.263, & 420 \text{ K} \leq T < 536 \text{ K} \\ -6.47 \times 10^{-4} T + 0.7663, & 536 \text{ K} < T \leq 800 \text{ K} \end{cases} \quad (9)$$

In order to reveal the transient response performances of porous volumetric solar receivers of all types, various materials for the porous skeleton and fluid are selected. High temperature alloys and refractory ceramics are commonly used in the solar radiation absorption at elevated temperature. The air is generally selected as heat transfer fluid for the porous volumetric solar receiver. Moreover, the molten salt is also proposed due to its spectrally selective absorption property which can limit the thermal radiation loss [9]. Therefore, the thermophysical properties of silicon carbide, 316 stainless steel, air, and Hitec molten salt are summarized in Table 1.

$$c_{p,\text{Air}} = 1.93 \times 10^{-10} T^4 - 8 \times 10^{-7} T^3 + 1.14 \times 10^{-3} T^2 - 4.49 \times 10^{-1} T + 1.06 \times 10^3 \quad (6)$$

4. Results and discussions

4.1. Transient response performances under the step change of solar flux

The initialization process of the porous volumetric solar receivers is simulated by applying a step increase in solar radiation from 0 to 1 MW/m². The outlet temperature of the heat transfer fluid and the average temperature of the porous structure are plotted over time, depicted in Fig. 3.

Noticeably, the outlet temperature responses differ significantly between the porous receivers using air and Hitec. The air receiver demonstrates a concave response curve, indicating an immediate temperature increase in response to the abrupt change in solar radiation. Conversely, the Hitec receiver exhibits a delayed response with an s-shaped curve. Quantitatively, the air receiver reaches its maximum

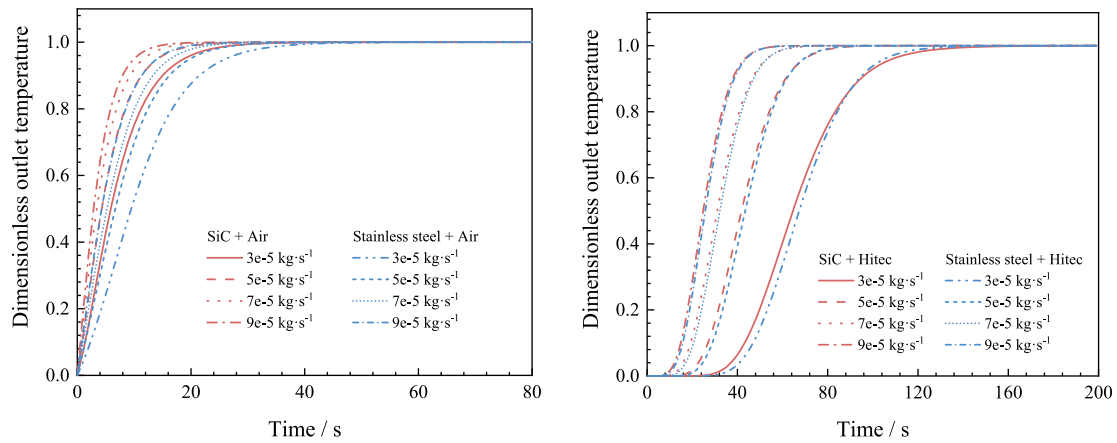


Fig. 6. Temperature variation under mass flow inlet of (a) air receivers (b) Hitec receivers.

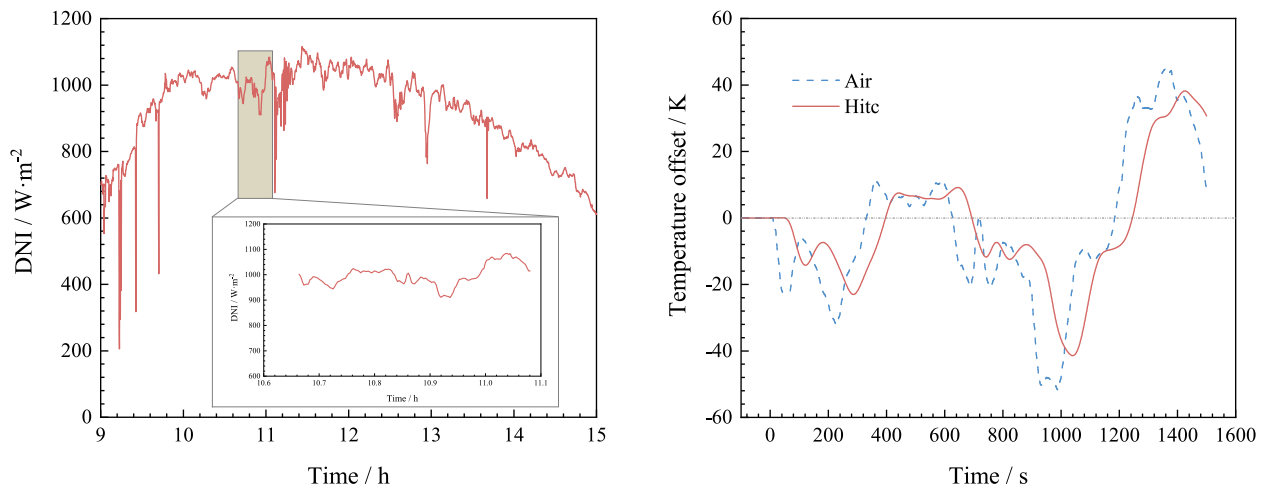


Fig. 7. (a) DNI variation (b) Temperature offset of air and Hitec porous receivers.

outlet temperature in about 20 s, whereas this duration extends to approximately 90 s for the Hitec receiver. The primary reason for this temperature response disparity is that the heat capacity of Hitec is significantly higher than that of air. The visual representation in Fig. 4 clarifies that the temperature field of the air receiver reaches steady state at 20 s compared to that at 60 s. However, the thermal stability of Hitec is not achieved at 20 s, and the overall fluid temperature still increases over time.

Significant temperature differences between the porous structure and air temperature are observable in Fig. 3(a) and Fig. 4(a). Due to the inefficiency of convective heat transfer, the temperature gap between outlet air and the porous surface reaches about 80.7 K upon reaching a steady state. In contrast, substituting air with Hitec enables the thermal equilibrium between the heat transfer fluid and the porous structure at the receiver outlet. Fig. 4(b) indicates a slight temperature difference at the rear of the Hitec receiver.

4.2. Parametric study of the transient response performance of the porous receivers

The transient behavior of the porous receivers is influenced by various factors, including the thermophysical properties and mass flow rate. The frequently utilized porous materials, such as silicon carbide and stainless steel, are applied to the porous structures. Both air and Hitec porous receivers under varying mass flow rates are analyzed. The time constant which is defined as the required time to reach 63.2 % of its steady state temperature is selected as the evaluation index for the receiver's transient response performance.

Fig. 5 exhibits a notable difference in time constants, indicating quicker responses of air receivers to solar radiation variation. Within the same inlet mass flow rate range, the average time constant for air receivers is 7.0 s, which significantly increases to 45.9 s for Hitec receivers. Furthermore, the fluid's mass flow rate influences the receiver's overall temperature, with lower mass flow rates resulting in higher fluid and

Table 2
Temperature offset of the air and Hitec porous receivers with and without PID control.

Materials	Maximum temperature offset K	Minimum temperature offset K	Standard deviation K
Air receiver without PID	44.8	−51.8	22.1
Air receiver with PID	5.9	−4.5	0.9
Hitec receiver without PID	38.2	−41.2	18.0
Hitec receiver with PID	34.1	−27.9	15.7

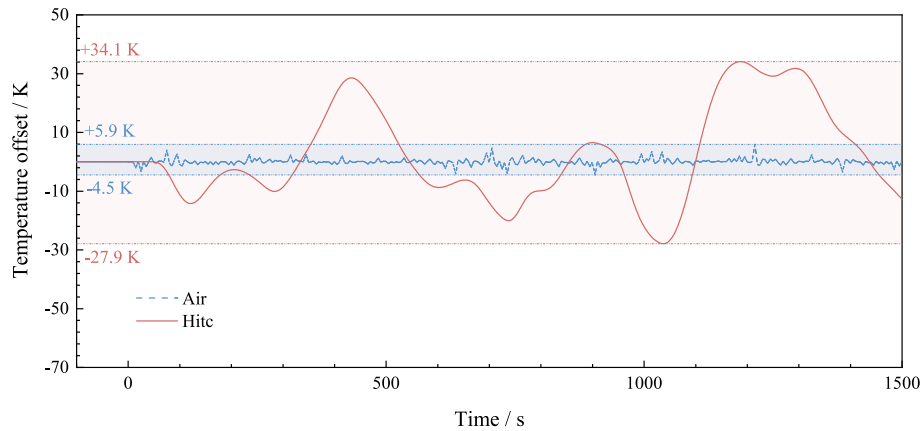


Fig. 8. Temperature offset of air and Hitec porous receivers with PID regulation.

porous skeleton temperatures. As the mass flow rate decreases, the temperature increasing rate becomes smaller. The time constant significantly increases by 129 % as the mass flow rate shifts from 9×10^{-5} kg/s to 3×10^{-5} kg/s. Moreover, the temperature variations over time under different mass flow rate are plotted in Fig. 6. It is demonstrated that the transient response curve shape is consistent for both air and Hitec receivers, with the mass flow rate's impact more evident in the Hitec system. Simulation results between silicon carbide and stainless steel show slight differences in the porous materials. The maximum difference resulting from porous material selection is found in the air receiver, with relative and absolute differences of 35.9 % and 4.4 s, respectively. It's noteworthy that the time constant in receivers with silicon carbide is consistently smaller due to its relatively smaller heat capacity compared to stainless steel.

4.3. PID regulation of porous volumetric solar receiver under variant incident flux

Under real situations, the variation of incident solar radiation for the receivers becomes more complex. Fig. 7(a) records the DNI variation within a day, and dramatic changes can be observed due to the cloud cover. DNI data spanning 1500 s is extracted and used to calculate the solar radiation intensity at the receiver inlet, as shown in the enclosed illustration of Fig. 7(a). The peak, lowest, and mean DNI values are 1084.0, 910.0, and 997.2 W/m², respectively.

The simulation of the outlet temperature for the porous receivers under actual solar radiation conditions is presented, and the temperature deviation from the steady-state temperature is illustrated in Fig. 7(b). Absent a regulation algorithm, the temperature fluctuation curves mirror the fluctuations in DNI, displaying consistent peaks and troughs in both DNI and temperature profiles. In comparison to the air receiver, a notable delay in temperature response to solar radiation is evident, resulting in an average delay of 56 s. Additionally, several minor crests and troughs in the air temperature profile are excluded due to the rapid changes in solar radiation and the delayed response of Hitec. Moreover, the magnitude of the temperature oscillation for both air and Hitec receivers is detailed in Table 2. It is noted that the temperature fluctuations are more evident in the air receiver.

Furthermore, the PID controller is applied to stabilize the outlet fluid temperature by the automatic regulation of the inlet mass flow rate for both air and Hitec porous receivers, and the temperature offsets are illustrated in Fig. 8. Evidently, the PID algorithm demonstrates effective control of the air receiver, showcasing only small variations with the maximum and minimum temperature offsets at 5.9 K and -4.5 K, respectively. However, regarding the Hitec receiver, the PID controller offers restricted regulation. Comparing the temperature profile in Fig. 7(b), it is noted that the values of temperature offset decrease, with the

maximum reduction reaching 13.3 K. Hence, more effective controlling algorithm such as feedforward-feedback control strategy may be necessary.

5. Conclusions

Transient response performances of different types of porous receivers under step change and actual solar radiation are investigated. The conclusions are derived as follows:

- (1) Air and molten salt porous receivers exhibit concave and s-shaped response curves, respectively, when subjected to a step increase in solar radiation. The higher heat capacity of molten salt results in a delayed response to solar radiation variation and prolongs the duration to reach a steady state.
- (2) The transient response of the porous receivers is significantly influenced by the heat transfer fluid's thermophysical properties and mass flow rate, while the effect of types of porous materials is not evident. The average time constant for air and molten salt receivers are 7.0 s and 45.9 s, respectively. Within the range of 9×10^{-5} to 3×10^{-5} kg/s, the lower mass flow rate increases the time constant by 129 %.
- (3) Temperature profiles follow the variation of actual solar radiation for both air and molten salt porous receivers, while a delay of 56 s is observed for the molten salt receiver. The PID algorithm can effectively reduce the amplitude of temperature fluctuation, particularly for the air receiver. The maximum temperature offsets for air and molten salt porous receivers are 5.9 K and 31.4 K, respectively.

CRediT authorship contribution statement

Shen Du: Writing – original draft, Visualization, Validation, Resources, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Ya-Ling He:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Meng-Jie Li:** Writing – review & editing, Visualization, Investigation, Formal analysis. **Zhan-Bin Liu:** Writing – review & editing, Validation, Software, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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